

InnovChain - Complete Concept

InnovChain

AI-Powered Blockchain Innovation Marketplace

The Problem

Every year, universities produce thousands of innovative projects that are never commercialized.

Current problems:

- Students don't know how to commercialize their work.
- Companies don't know these projects exist.
- Universities have no way to track innovation impact.
- Students fear idea theft.
- Licensing agreements are slow and rely on trust.

Proposed Solution

A decentralized innovation marketplace where:

- Students publish projects
- Universities verify authenticity
- Companies discover solutions
- AI recommends the best matches
- Blockchain secures ownership and licensing

Instead of storing everything on-chain, only proof of ownership and agreements are stored on the blockchain.

Main Users

Student

- Upload project
- Choose visibility
- Generate ownership proof
- Accept licensing offers
- Track royalties
- Chat with companies

Company

- Search projects
- Post innovation challenges
- Request demonstrations
- Buy licenses
- Hire students

University

- Verify projects
- Certify student identity
- Validate originality
- Approve publications

Admin

- Manage users
- Moderate content
- Resolve disputes

Architecture

React + Tailwind

↓
Node.js / Express
↓
MongoDB
↓
AI Recommendation Engine
↓
IPFS (Project Storage)
↓
Blockchain (Cardano / Ethereum / Hedera)
↓
Smart Contracts

Blockchain Usage
Project.pdf

↓
SHA-256 Hash
↓
Store hash on blockchain

Blockchain stores:

- Project hash
- Timestamp
- Student wallet
- University signature
- License agreements
- Royalty records

Smart Contracts

Ownership Contract:

Owner
Project ID
Hash
Timestamp

Licensing Contract:

Company requests:

6 months

Commercial use

1000\$

Student accepts.

The smart contract locks payment, creates the agreement and transfers the license automatically.

Royalty Contract:

10% Student

5% University

85% Company

AI Recommendation System

Use semantic embeddings instead of keyword search.

Example:

Student project: AI-powered irrigation system

Company request: Agriculture automation

Technologies:

- Sentence Transformers
- OpenAI Embeddings
- FAISS
- ChromaDB

Recommendation Score: 89%

AI Features

1. Project Recommendation
2. Company Recommendation
3. Similar Projects
4. Automatic Tags
5. Project Summarization
6. Innovation Score
7. Duplicate Detection

Automatic tags:

Blockchain

Healthcare

AI

IoT

Cybersecurity

Innovation Score evaluates:

- Novelty
- Sustainability
- Market Potential
- Technical Complexity

Duplicate Detection:

92% Similar

Sustainability

Projects receive SDG labels:

SDG 3 Health

SDG 7 Clean Energy

SDG 13 Climate Action

Search Filters

AI, Blockchain, IoT, Cybersecurity, Healthcare, Agriculture, Energy, Sustainability, University, Year, Programming Language, Patent Status, License Type.

Dashboards

Student Dashboard:

Views, Interested Companies, Pending Offers, Earnings, Royalties, Certificates.

Company Dashboard:

Recommended Projects, Saved Projects, Active Licenses, Students Contacted, Hiring Pipeline.

University Dashboard:

Published Projects, Commercialized Projects, Revenue, Research Impact, Innovation Statistics.

Security Features

JWT Authentication

MFA
Wallet Authentication
RBAC
File Encryption
Malware Scanning
Audit Logs
Secure Uploads
Rate Limiting
Input Validation
HTTPS
OWASP Protections

Extra Features
Patent Readiness Checker
Pitch Generator
Collaboration with ownership percentages
Hackathon Integration
Innovation Challenges
Soulbound NFT Certificate

Technology Stack

Frontend:

React, TypeScript, Tailwind CSS, React Router, Redux Toolkit

Backend:

Node.js, Express, MongoDB

AI:

Python, FastAPI, Sentence Transformers, FAISS, LangChain (optional)

Blockchain:

Cardano / Hedera / Ethereum

Storage:

MongoDB + IPFS + Blockchain

Authentication:

JWT, Wallet Authentication, OAuth

Why this is a strong final-year project

It combines Full-Stack Web Development, Artificial Intelligence, Blockchain, Cybersecurity and Sustainability into one complete platform capable of solving a real-world problem with industrial relevance.